江苏省苏州中学 国际部
Suzhou High School – International Division

SZZX-ID
Bridge Program
Physics

Unit 1 Lesson 3: suvat equations and freefall

Class Outline

Title of Unit	Linear motion
Title of Lesson	Suvat equations and freefall
Starter Activity	Uniform motion equation – revision
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	1. Derivation of first suvat equation 2. Lecture: steps to solve a kinematic problem 3. Inquiry: freefall motion
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	1. Vocab before lesson 2. Repeat and speak slowly when delivering content 3. Encourage teamwork for acceleration sample question
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: 1. Understand how the equations of motion was derived from the definitional of acceleration 2. Apply suvat equations in problem solving 3. Understanding that freefall is a special UAM 4. Know the concept of terminal velocity
ATLs	Identify unit ATLs and how they are being addressed in the lesson
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Communicators: Using terminology, symbols and communication conventions consistently and correctly
Teaching Activities	Sample questions Video: freefall on the moon
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Homework reminder – submit hardcopy; read the instructions

Vocabulary

- Derive/derivation 推导
- Brake 刹车
- Freefall 自由落体
- Gravity 重力
- Free-fall acceleration/gravitational acceleration 重力加速度
- Incline 斜坡
- Leaning tower of Pisa 比萨斜塔
- Uniform 一致的
- Air drag/air resistance 空气阻力
- Latitude 纬度
- Altitude 高度

Starting from the definition of acceleration

- Acceleration is the rate of change of velocity

$$a = \frac{\Delta v}{\Delta t}$$

- For uniformly accelerated motion:

$$a = \frac{v-u}{t} \Rightarrow \boxed{v = u + at} \quad (1)$$

Where *v* and *u* are the final and initial velocity, t is the interval of the motion

Suvat equations – derivation

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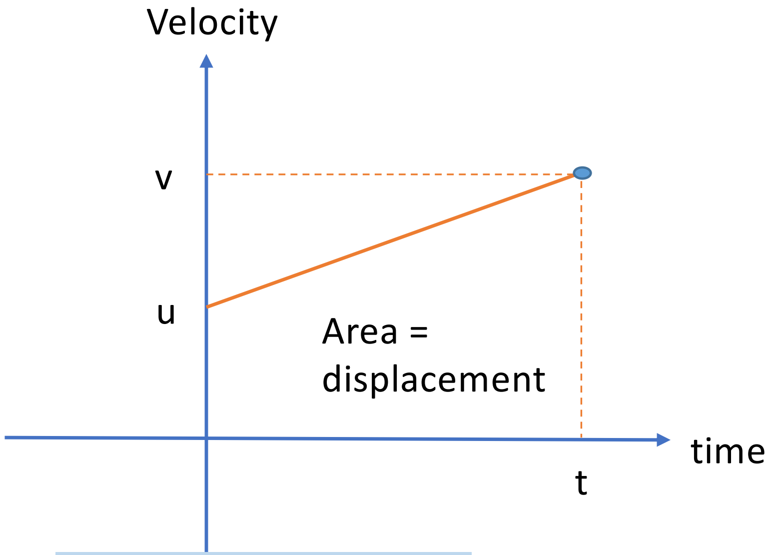

$$s = \frac{(u+v)\times t}{2}$$ ②

S: displacement

In ②, substitute *v* with $v = u + at$

$$s = \frac{(u+u+at)\times t}{2}$$

$$= ut + \frac{1}{2}at^2$$ ③

In ②, substitute *t* with $t = \frac{v-u}{a}$

$$s = \frac{(v+u)(v-u)}{2a}$$

So:

$$v^2 - u^2 = 2as$$ ④

Conditions for suvat equations to be applied

The equations of motion ONLY APPLY when:

- The motion is in a straight line
- There is constant acceleration

1 $$v = u + at$$

2 $$s = \left(\frac{v+u}{2}\right)t$$

3 $$s = ut + \frac{1}{2}at^2$$

4 $$v^2 = u^2 + 2as$$

Example 1

A cyclist is cycling at 2.4 m s^{-1}.

They accelerate for 7.0 s and

reach a velocity of 3.9 m s^{-1}.

Calculate the distance

travelled in the 7.0 s.

Solution steps	Calculations
Step 1: Write out the values given in the question and convert the values to the units required for the equation.	$u = 2.4\ \mathrm{m\ s^{-1}}$ $v = 3.9\ \mathrm{m\ s^{-2}}$ $t = 7.0\ \mathrm{s}$
Step 2: Write out the equation.	$s = \left[\frac{(u+v)}{2}\right]t$
Step 3: Substitute the values given.	$= \left[\frac{(2.4+3.9)}{2}\right] \times 7.0$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding	$= 22.05\ \mathrm{m} = 22\ \mathrm{m}\ (2\ \mathrm{s.f.})$

Example 2

A car travels with velocity of 20m s^{-1}. It brakes and slows down with an acceleration of 4.0m $s^{-2.}$.

- How far has it travelled when the velocity is dropped to 8m s^{-1}
- How far would it travel in 10 seconds after it braked?

RULE 1: Always draw a sketch before starting.
RULE 2: Choose which direction is positive
RULE 3: List your quantities
RULE 4: Choose the correct equation and substitute in the correct quantities.

Example 3

An object travels with an initial velocity of 20 m s^{-1}, with a constant acceleration of –5 m s^{-2}

- What will its displacement be after 20 s?

- How does the *v*-t graph look like?

Freefall motion- From Aristotle to Galileo

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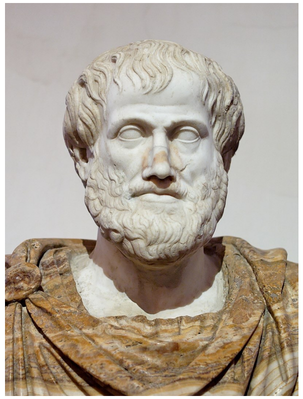
- Aristotle (384 – 322 B.C) has his view on freefall: an object’s mass affected the rate that it would hit the ground (heavier objects fell faster than lighter ones)

- Galileo (1564-1642) developed a different view
  - an object’s mass does not affect the rate it falls (two objects with different masses dropped at the same time will reach the ground at the same time)
  - Logic and evidence of Galileo includes:

◆ Thought experiment (a big rock and a small rock chained)
◆ Experiments of objects fell down inclined planes to “dilute” gravity
◆ The famous (likely fictional) experiment on the Leaning Tower of Pisa



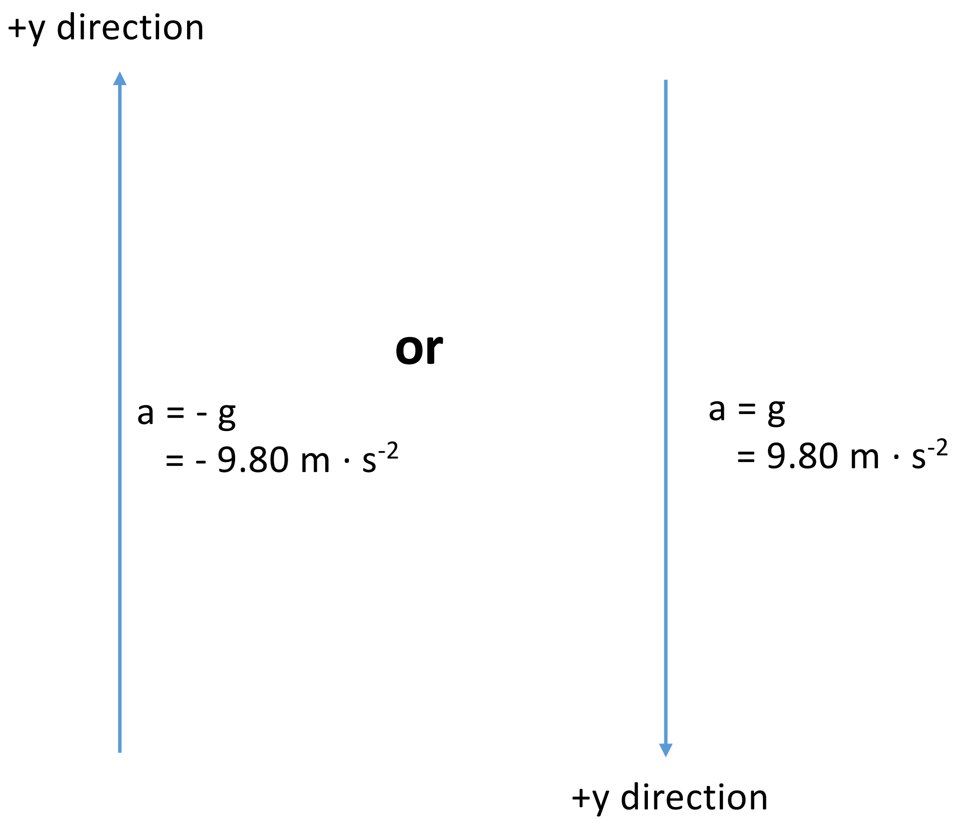
Pictures from:
https://en.wikipedia.org/wiki

Free fall motion - a special UAM

- A freely falling object is any object moving freely under the influence of gravity only, regardless of its initial motion
- The magnitude of the free-fall acceleration is denoted by the symbol ***g***
- At a given location on the Earth and in the absence of air resistance, all objects fall with the same uniform acceleration.
- At Earth's surface, the value of g is approximately $9.80\ m \cdot s^{-2}$

(some most frequently used approximate value of g:

$9.80\ m \cdot s^{-2}$, $9.81\ m \cdot s^{-2}$, $10\ m \cdot s^{-2}$)



Example 1

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Image from shutterstock.com

A construction worker accidentally drops a brick from a high scaffold 10 m above ground.

Determine:

a. The time taken for the brick to hit the ground
b. The final velocity of the brick when hitting the ground

(take $g = 9.8\ ms^{-2}$ as found in Data Booklet. Ignore air friction)

Example 2

A raindrop falls from the sky 4500 m above ground. What is the velocity when the raindrop reaches the ground?

300 ms^{-1} (!)
That's the velocity of a muzzle bullet
Then are we alive when walking in the rain?

In reality average velocity of raindrops is about 9ms^{-1}

What factor is causing the discrepancy?

Terminal velocity

- When air drag cannot be ignored， the falling object will reach a final constant velocity rather than accelerating all the way
  - If the object is quite light (e.g. a feather)
  - When the object is falling from a great height
  - This final velocity is called terminal velocity